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ROUND-TRIP RATE FOR NRPT

- Hence the round trip rate satisfies:

$$\tau_{\text{NRPT}} = \frac{1}{2 + 2 \sum_{n=1}^N \frac{r_n}{1 - r_n}}$$



► **The first term is the minimum round trip time if taking a direct path $2(N+1)$**

▶ The second term is the additional expected time due to rejections (same as RPT)

$$\geq \frac{1}{2N + 2 \sum_{n=1}^N \frac{r_n}{1 - r_n}} = \tau_{\text{RPT}}$$

► Moreover as $N \rightarrow \infty$ and $\max |\beta_{n-1} - \beta_n| \rightarrow 0$ there exists $\Lambda > 0$, such that

$$\lim_{N \rightarrow \infty} \tau_{\text{NRPT}} = \frac{1}{2 + 2\Lambda} > 0$$

Scalable to GPUs' improves performance increased parallelism (with marginal gains)

► **Theorem:** Let $r_n = s_{n,-}$ be the rejection probability for the swap between chains $n-1$ and n

- The expected round trip time equals:

$$\mathbb{E}_{\text{NRPT}}[\Delta T] = 2(N+1) + 2(N+1) \sum_{n=1}^N \frac{r_n}{1-r_n}$$

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- ▶ The expected round trip time equals:

$$\mathbb{E}_{\text{NRPT}}[\Delta T] = 2(N + 1) + 2(N + 1) \sum_{n=1}^N \frac{r_n}{1 - r_n}$$

- ▶ Hence the round trip rate satisfies:

$$\tau_{\text{NRPT}} = \frac{1}{2 + 2 \sum_{n=1}^N \frac{r_n}{1 - r_n}} \geq \frac{1}{2N + 2 \sum_{n=1}^N \frac{r_n}{1 - r_n}} = \tau_{\text{RPT}}$$

- ▶ Moreover as $N \rightarrow \infty$ and $\max |\beta_{n-1} - \beta_n| \rightarrow 0$ there exists $\Lambda > 0$, such that

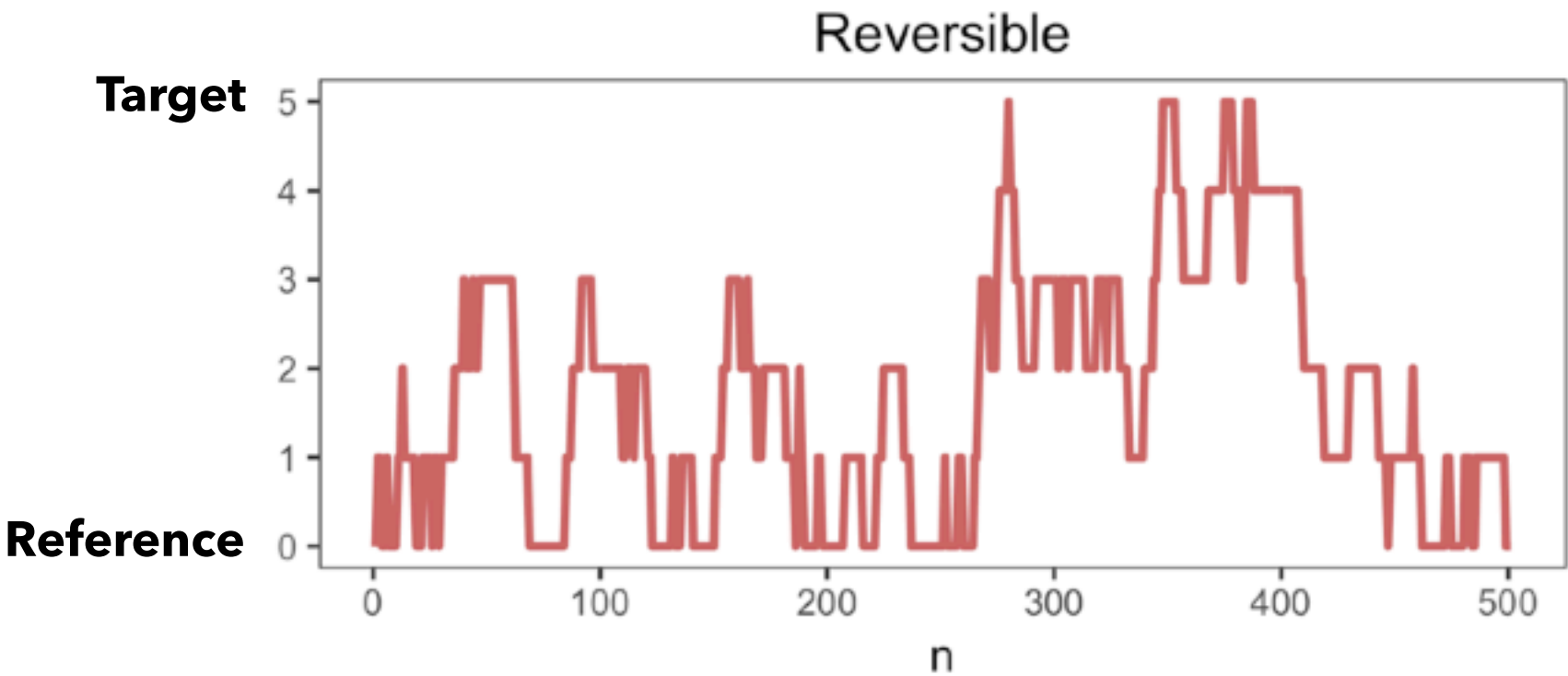
$$\lim_{N \rightarrow \infty} \tau_{\text{NRPT}} = \frac{1}{2 + 2\Lambda} > 0$$

- ▶ **The first term** is the minimum round trip time if taking a direct path $2(N + 1)$
- ▶ **The second term** is the additional expected time due to rejections (same as RPT)
- ▶ Non-asymptotically dominates RPT
- ▶ Scales to GPUs' improves in performance increased parallelism (with marginal gains)

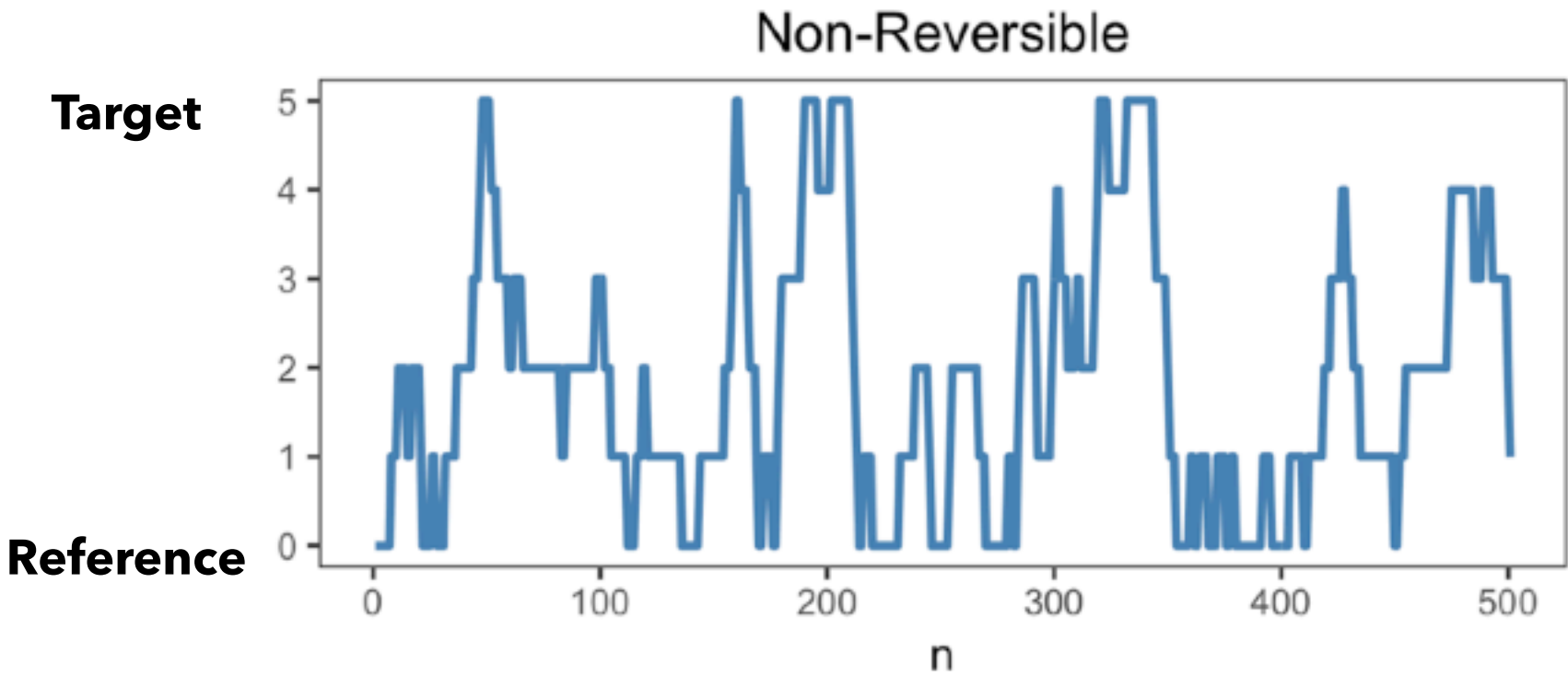
SAMPLE TRAJECTORIES

$$N = 5$$

Round Trips



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